

Unifying General Relativity and Quantum Mechanics through Triadic Closure

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Abstract

General Relativity and quantum mechanics are individually complete yet architecturally incompatible: GR describes spacetime as continuous geometric structure, while QM describes physical evolution through informational and energetic constraints. A century of unification attempts—including string theory, loop quantum gravity, emergent-spacetime programs, and non-local extensions—has not resolved this incompatibility, largely because these approaches introduce new microscopic ingredients rather than addressing the architectural mismatch between geometric and informational roles.

This paper presents a minimal and falsifiable unification of GR and QM based on *triadic closure*, the structural mechanism developed across the Triadic Cosmos ecosystem. Earlier work established energy, geometry, and information as three irreducible roles instantiated across modern physics; formalized their closed interdependence through the Triadic Cycle; embedded triadic recursion within holographic renormalization; and introduced local closure cosmology as a structural explanation for the absence of new fundamental physics. Together, these works revealed that physical reality is organized by horizon-bounded, triadically recursive closures whose emergent temporal structure arises from successive triadic updates.

The present paper consolidates these insights into a single architectural mechanism that reconciles GR and QM without new fields, dimensions, symmetries, or dynamical extensions. Triadic closure provides the minimal structure through which geometric, energetic, and informational roles cohere; explains why GR is overcomplete and why only a subset of its solutions are physically realizable; resolves the measurement and non-locality paradoxes of QM; and yields a unified account of emergent time, causal structure, and horizon dynamics. Within each closure, GR and QM become two complementary expressions of the same triadic architecture.

By identifying triadic closure as the structural origin of physical law, this paper positions the Triadic Cosmos as a complete and falsifiable unification framework. It shows that modern physics already contains all necessary ingredients for unification, and that the apparent fragmentation between GR and QM arises not from missing physics but from the absence of an architectural grammar linking their roles. Triadic closure supplies that grammar, yielding a minimal, coherent, and observationally grounded unification of general relativity and quantum mechanics.

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1 Introduction

General Relativity (GR) and quantum mechanics (QM) form the two foundational pillars of modern physics. GR describes spacetime as a continuous geometric structure governed by the Einstein field equations, while QM describes physical evolution through informational and energetic constraints encoded in Hilbert-space dynamics. Each framework is individually complete, precise, and empirically validated, yet their architectural assumptions are mutually incompatible. GR treats geometry as fundamental and time as a geometric ordering; QM treats information as fundamental and time as a parameter governing unitary evolution. A century of unification attempts has not resolved this mismatch. Instead, major programs—including string theory, loop quantum gravity, causal set theory, emergent-spacetime proposals, and non-local extensions—have introduced new microscopic ingredients without providing a minimal and falsifiable architectural reconciliation of geometric and informational roles.

The Triadic Cosmos ecosystem approaches this longstanding problem from a different direction. Earlier work introduced a minimal ontology consisting of three irreducible structural roles: energy, geometry, and information Witte 2026c. These roles form a closed interdependence expressed through the Triadic Cycle $E \rightarrow G \rightarrow I \rightarrow E$ Witte 2026d. Subsequent papers derived theory-independent architectural constraints required for physical realizability Witte 2026b, extended the triadic structure into recursive and fractal domains Witte 2026f, embedded triadic recursion within holographic renormalization Witte 2026e, and introduced local closure cosmology as a structural explanation for the absence of new fundamental physics across cosmology, high-energy experimentation, computation, and theoretical research Witte 2026a. Together, these works revealed that physical reality is organized by horizon-bounded, triadically recursive closures whose emergent temporal structure arises from successive triadic updates.

The present paper consolidates these insights into a single architectural mechanism—*triadic closure*—that reconciles GR and QM without new fields, dimensions, symmetries, or dynamical extensions. Triadic closure provides the minimal structure through which geometric, energetic, and informational roles cohere. Within each closure, GR and QM become two complementary expressions of the same triadic architecture: GR describes the geometric constraints that shape informational capacity, while QM describes the informational and energetic constraints that shape geometric evolution. Time emerges from recursion rather than from geometry or unitarity alone, resolving the temporal mismatch between the two theories. Causal structure becomes horizon-bounded and closure-dependent, eliminating non-locality paradoxes and measurement inconsistencies. GR’s overcomplete solution space is filtered by triadic closure, explaining why only a subset of geometries are physically realizable. QM’s informational dynamics become embedded within geometric constraints, resolving the measurement problem and the apparent tension between entanglement and relativistic causality.

By identifying triadic closure as the structural origin of physical law, this paper positions the Triadic Cosmos as a complete and falsifiable unification framework. It shows that modern physics already contains all necessary ingredients for unification, and that the apparent fragmentation between GR and QM arises not from missing physics but from the absence of an architectural grammar linking their roles. Triadic closure supplies that grammar, yielding a minimal, coherent, and observationally grounded unification of general relativity and quantum mechanics.

2 Related Work

Unification efforts in modern physics span several major traditions, each illuminating different structural aspects of physical law. The present work differs from these approaches by introducing *triadic closure* as a minimal architectural mechanism capable of reconciling general relativity (GR) and quantum mechanics (QM) without new fields, dimensions, or dynamical

extensions. This section situates triadic closure within the broader landscape of gravitational theory, quantum information, holography, emergent-time programs, and cosmology, and clarifies how the Triadic Cosmos ecosystem provides the architectural foundation for the present unification proposal.

2.1 General Relativity, Black-Hole Thermodynamics, and Overcomplete Geometry

General Relativity provides the geometric backbone of modern physics, describing curvature through the Einstein field equations Einstein 1915. Black-hole thermodynamics revealed that geometric area encodes informational capacity, with Bekenstein’s entropy bound Bekenstein 1973 and Hawking radiation Hawking 1975 establishing a deep link between geometry, energy, and information. These results demonstrate that GR’s geometric degrees of freedom implicitly carry informational structure, but GR itself does not constrain informational embedding or recursive evolution.

Earlier work in the Triadic Cosmos program showed that GR is an *overcomplete architectural model*: it admits geometries that lack informational or energetic roles and therefore cannot instantiate full triadic closure Witte 2026c. This overcompleteness motivates the need for an architectural filter capable of distinguishing physically realizable geometries from mathematically permissible ones.

2.2 Holography, AdS/CFT, and Geometry–Information Duality

The holographic principle Hooft 1993; Susskind 1995 and the AdS/CFT correspondence Maldacena 1998; Witten 1998 provide the most explicit realization of geometry–information interdependence. The Ryu–Takayanagi formula and its covariant extensions Ryu and Takayanagi 2006; Hubeny, Rangamani, and Takayanagi 2007 relate entanglement entropy to extremal surfaces, while tensor-network and quantum-error-correcting code models Pastawski et al. 2015; Raamsdonk 2010 show how bulk geometry can emerge from boundary informational structure.

These developments instantiate the $G \rightarrow I$ and $I \rightarrow G$ legs of the triadic cycle and demonstrate that informational structure constrains geometric connectivity. Earlier work in the Triadic Cosmos ecosystem embedded triadic recursion within holographic renormalization, interpreting the recursion index as a discrete RG direction and showing that AdS/CFT provides a canonical—but not necessary—realization of triadic interdependence Witte 2026e. The present paper extends this architectural perspective by showing that triadic closure reconciles GR and QM even in the absence of global holographic dualities.

2.3 Quantum Information, Computational Limits, and Emergent Time

Information-theoretic approaches to physics—from Wheeler’s “it from bit” to modern quantum information theory Nielsen and Chuang 2010—highlight the structural role of informational capacity in physical evolution. Quantum speed limits Margolus and Levitin 1998; Mandelstam and Tamm 1945 and computational bounds Lloyd 2000 link energetic resources to the rate of informational change, while the Page–Wootters mechanism Page and Wootters 1983 interprets temporal ordering as emerging from correlations rather than from a fundamental time parameter.

The Triadic Cycle formalizes these insights by treating time as the ordering induced by successive triadic updates Witte 2026d. This emergent-time interpretation is compatible with both relativistic time dilation and quantum unitary evolution, providing a shared temporal structure for GR and QM. The present work builds on this foundation by showing that emergent time is closure-dependent: each observer occupies a horizon-bounded dynamical niche with its own recursion index Witte 2026a.

2.4 Quantum Gravity, Relational Ontologies, and Structural Approaches

Quantum gravity programs such as loop quantum gravity Rovelli 2004, spin-foam models Perez 2013, and causal set theory Bombelli et al. 1987 explore discrete or relational realizations of geometry. Structural realist and relational ontologies Ladyman and Ross 2007; Rovelli 1996 argue that invariant relational structure may be more fundamental than objects or fields. These approaches resonate with the triadic ontology, which identifies energy, geometry, and information as irreducible structural roles.

However, none of these frameworks provide a closed architectural mechanism linking GR and QM. The Triadic Cosmos ecosystem introduced such a mechanism by identifying triadic closure as the minimal interdependence required for physical realizability Witte 2026b; Witte 2026f. The present paper extends this architectural foundation into a full unification proposal.

2.5 Cosmology, JWST, and Closure-Dependent Parameters

Cosmological observation has entered a high-precision era, with JWST revealing early-universe structure at unprecedented resolution J. Collaboration 2023. The persistent Hubble tension—disagreement between early-time global measurements and late-time local measurements Riess et al. 2022; P. Collaboration 2020—has motivated speculation about new physics.

Local closure cosmology provides an alternative interpretation: early-time and late-time measurements probe distinct closure regimes with different recursion indices and horizon histories Witte 2026a. This reframes cosmological parameters as closure-dependent rather than universal and explains why no new fundamental physics has emerged despite increasing observational reach.

2.6 The Triadic Cosmos Ecosystem and Architectural Unification

The present work builds directly on the Triadic Cosmos ecosystem, which introduced the triadic ontology Witte 2026c, formalized the triadic cycle Witte 2026d, derived architectural constraints for physical unification Witte 2026b, extended the framework into recursive and fractal domains Witte 2026f, embedded triadic recursion within holographic renormalization Witte 2026e, and developed local closure cosmology as a structural explanation for the absence of new physics Witte 2026a. Together, these works provide the architectural backbone for the present unification proposal.

The contribution of this paper is to consolidate these insights into a single structural mechanism—*triadic closure*—that reconciles GR and QM without new fields, dimensions, or dynamical extensions. Triadic closure provides the minimal architecture through which geometric, energetic, and informational roles cohere, and thereby supplies a unified foundation for modern physics.

3 Triadic Closure as Unification Mechanism

Triadic closure is the minimal architectural mechanism through which the structural roles of energy, geometry, and information cohere into a single unified physical framework. It provides the conceptual and mathematical foundation that reconciles general relativity and quantum mechanics without introducing new fields, dimensions, symmetries, or dynamical extensions. In this section we formalize triadic closure, describe its structural consequences, and show how it resolves the longstanding incompatibilities between GR and QM.

3.1 The Triadic Roles

Earlier work in the Triadic Cosmos ecosystem identified three irreducible structural roles underlying all physical theories: energy (E), geometry (G), and information (I) Witte 2026c. These

roles are not microscopic variables but architectural constraints that govern the organization of physical law. Each role is indispensable:

- E determines dynamical resources, causal influence, and stress–energy content.
- G determines causal structure, horizon formation, and geometric connectivity.
- I determines distinguishability, entropy, entanglement, and informational capacity.

The mutual interdependence of these roles is expressed through the Triadic Cycle $E \rightarrow G \rightarrow I \rightarrow E$, which formalizes how each role reorganizes the others across successive dynamical updates Witte 2026d. Triadic closure occurs when this cycle is well-defined, complete, and recursively stable.

3.2 The Recursive Update Operator

The Triadic Cycle becomes dynamical when expressed through the recursive update operator R , defined by

$$X_{n+1} = R(X_n), \quad X_n = (E_n, G_n, I_n).$$

Each update reorganizes the triadic roles according to their functional dependencies:

$$E_{n+1} = f(G_n, I_n), \quad G_{n+1} = g(E_n, I_n), \quad I_{n+1} = h(E_n, G_n).$$

This recursion induces an ordering on successive triadic states. Temporal structure is therefore emergent rather than fundamental: time is the ordering of $\{X_0, X_1, X_2, \dots\}$. This interpretation is compatible with both relativistic and quantum notions of time and provides a shared temporal architecture for GR and QM.

3.3 Definition of Triadic Closure

A physical structure exhibits *triadic closure* if and only if:

1. all three roles satisfy positivity constraints ($E > 0, G > 0, I > 0$),
2. informational embedding is holographically consistent,
3. energetic and geometric roles satisfy causal accessibility,
4. the recursive update operator R is well-defined,
5. the full cycle $E \rightarrow G \rightarrow I \rightarrow E$ closes without contradiction.

If any leg of the cycle fails, recursion becomes undefined, emergent time cannot arise, and the structure cannot correspond to physical reality Witte 2026b. Triadic closure therefore acts as an architectural filter that distinguishes physically realizable configurations from merely mathematically permissible ones.

3.4 GR as Overcomplete Geometry and QM as Overcomplete Information

GR is intentionally broad: the Einstein equations constrain curvature but do not restrict informational capacity, holographic embedding, or recursive evolution. As a result, GR admits geometries that lack informational or energetic roles, including white holes, naked singularities, traversable wormholes, closed timelike curves, and exotic compact objects. These geometries are mathematically valid but architecturally incomplete Witte 2026c.

QM and QFT are likewise overcomplete: Hilbert-space dynamics allow informational structures that cannot be embedded in consistent geometric or causal frameworks. Triadic closure filters both theories simultaneously, selecting the subset of GR geometries and QM informational structures that can coexist within a single recursive architecture.

3.5 Emergent Time and Unified Causality

Triadic closure resolves the temporal mismatch between GR and QM. In GR, time is geometric: it arises from spacetime curvature and causal structure. In QM, time is parametric: it indexes unitary evolution. Triadic recursion provides a shared origin for both:

- geometric time arises from the $E \rightarrow G$ update,
- quantum time arises from the $I \rightarrow E$ update,
- emergent time arises from the ordering of recursive triadic states.

Causality becomes horizon-bounded and closure-dependent, eliminating non-locality paradoxes and reconciling entanglement with relativistic causal structure.

3.6 Triadic Closure as the Unification of GR and QM

Within each closure, GR and QM become two complementary expressions of the same triadic architecture:

- GR describes how energetic and informational roles shape geometric structure.
- QM describes how geometric and informational roles constrain energetic evolution.
- holography describes how geometric structure encodes informational capacity.
- thermodynamics describes how informational change generates energetic structure.

Triadic closure therefore provides the minimal architectural grammar linking GR and QM. It explains why no new fundamental physics has been discovered, why GR's exotic solutions are excluded, why QM's non-locality paradoxes dissolve, and why cosmological parameters become closure-dependent rather than universal.

In this sense, triadic closure is not an extension of GR or QM but the structural mechanism that reveals their coherence. It supplies the architectural foundation through which general relativity and quantum mechanics unify into a single, minimal, and falsifiable physical framework.

4 GR and QM as Complementary Triadic Roles

General Relativity and quantum mechanics are traditionally viewed as conceptually incompatible: GR treats geometry as fundamental and continuous, while QM treats information and energetic constraints as fundamental and discrete. Triadic closure reveals that this incompatibility arises not from the theories themselves, but from the absence of an architectural grammar linking their structural roles. When energy, geometry, and information are treated as irreducible and mutually interdependent, GR and QM emerge as complementary realizations of the same triadic architecture.

4.1 Geometry as the Constraint Layer (GR)

In GR, geometry determines causal structure, horizon formation, and the accessibility of physical processes. The Einstein field equations relate curvature to energetic content, but do not constrain informational capacity or recursive evolution. Within triadic closure, geometry (G) plays the role of the *constraint layer*:

- geometric curvature bounds informational capacity through holographic area laws,
- causal structure determines which informational updates are accessible,
- horizon formation defines the boundary of a closure and its recursion index.

GR therefore realizes the $E \rightarrow G$ leg of the triadic cycle: energetic structure shapes geometric constraints, which in turn regulate informational embedding.

4.2 Information as the Generative Layer (QM)

Quantum mechanics describes physical evolution through informational and energetic constraints encoded in Hilbert-space dynamics. Entanglement, distinguishability, entropy, and measurement all reflect informational structure. Within triadic closure, information (I) plays the role of the *generative layer*:

- informational structure determines energetic evolution through unitarity,
- entanglement generates geometric connectivity (EP=EPR),
- informational change drives energetic reconfiguration (Landauer-type relations).

QM therefore realizes the $I \rightarrow E$ leg of the triadic cycle: informational structure generates energetic structure, which in turn shapes geometric evolution.

4.3 Energy as the Dynamical Layer (QFT and Thermodynamics)

Energy (E) mediates between geometry and information. In QFT, energetic excitations propagate through geometric structure; in thermodynamics, informational change generates energetic cost; in holography, boundary energy sources bulk geometry. Within triadic closure, energy plays the role of the *dynamical layer*:

- energetic content determines curvature and causal accessibility,
- energetic processes generate or reorganize informational structure,
- energetic constraints regulate the rate of recursive updates (quantum speed limits).

Energy therefore realizes the $G \rightarrow I$ leg of the triadic cycle: geometric structure bounds informational capacity, which in turn constrains energetic evolution.

4.4 Complementarity Through Triadic Closure

When the triadic roles are considered jointly, GR and QM become complementary rather than conflicting:

- GR provides the geometric constraints (G) that bound informational capacity (I).
- QM provides the informational structure (I) that generates energetic evolution (E).

- QFT and thermodynamics provide the energetic dynamics (E) that shape geometry (G).

The full cycle closes:

$$E \rightarrow G \rightarrow I \rightarrow E,$$

yielding a unified architecture in which GR and QM are not competing descriptions but structural complements. GR describes the constraint layer of physical reality; QM describes the generative layer; QFT and thermodynamics describe the dynamical layer. Triadic closure is the mechanism through which these layers cohere.

4.5 Unified Temporal and Causal Structure

Triadic closure also reconciles the temporal and causal assumptions of GR and QM. In GR, time is geometric; in QM, time is parametric; in triadic closure, time is emergent from recursion. Causality becomes horizon-bounded and closure-dependent, eliminating the apparent tension between entanglement and relativistic causal structure. EP=EPR becomes a direct expression of the $G \leftrightarrow I$ interdependence.

4.6 Summary

GR and QM are not architecturally incompatible. They are two complementary realizations of the same triadic structure: geometry as constraint, information as generator, and energy as dynamics. Triadic closure provides the minimal architectural grammar linking these roles, yielding a unified and falsifiable foundation for modern physics.

5 Revising the Triadic Cosmos Unification: From Global Dualities to Local Triadic Closure

The Triadic Cosmos: A Foundation for Unification Witte 2026c introduced the first complete architectural synthesis of modern physics, identifying energy, geometry, information, and recursive update as the structural roles through which General Relativity (GR), quantum field theory (QFT), the Standard Model (SM), holography, EP=EPR, renormalization group (RG) flow, and black-hole thermodynamics align into a single conceptual framework. In that formulation, AdS/CFT played a central role as the canonical dynamical realization of triadic recursion and as the clearest example of geometry–information interdependence. The present work revises and strengthens that unification by showing that AdS/CFT is not a necessary ingredient of the triadic architecture. Instead, triadic closure provides a fully local, horizon-bounded, and theory-neutral mechanism through which GR and QM unify without requiring global dualities, extra dimensions, or conformal boundary structure.

5.1 Limitations of Global Holographic Dependence

In the original Triadic Cosmos formulation, holography and AdS/CFT served as the primary instantiation of the $G \rightarrow I$ and $I \rightarrow G$ legs of the triadic cycle. While this mapping is structurally elegant, it relies on assumptions that are not universally realized in physical spacetime:

- the existence of a global conformal boundary,
- asymptotically AdS geometry,
- conformal symmetry of boundary degrees of freedom,
- global RG flow as a proxy for emergent time.

These assumptions are mathematically powerful but cosmologically restrictive. They do not apply to our universe, which is not asymptotically AdS, does not possess a global conformal boundary, and exhibits horizon-bounded causal structure rather than global holographic accessibility. As a result, AdS/CFT can serve as an illustrative special case of triadic interdependence, but not as a universal foundation for unification.

5.2 Local Triadic Closure as the Correct Architectural Mechanism

Local closure cosmology Witte 2026a revealed that physical reality is organized not by global dualities but by *local triadic closures*: horizon-bounded dynamical niches defined by causal structure, informational capacity, and recursion index. Within each closure:

- geometry bounds informational capacity through horizon area,
- informational structure constrains energetic evolution,
- energetic structure shapes geometric curvature,
- recursion generates emergent time.

This mechanism is fully local, requires no global boundary, and applies to all physically realized spacetimes. It therefore replaces AdS/CFT as the universal dynamical realization of the triadic cycle. Holography becomes a special case of closure-dependent informational embedding rather than a global requirement for unification.

5.3 Strengthening the Original Unification

The present work strengthens the unification proposed in Witte 2026c in three ways:

1. Removing non-essential theoretical dependencies. AdS/CFT, conformal symmetry, and global holographic dualities are no longer required for the triadic architecture. Triadic closure provides a minimal and theory-neutral mechanism that applies equally to GR, QM, QFT, thermodynamics, and cosmology.

2. Replacing global recursion with local emergent time. In the original formulation, holographic RG flow served as the dynamical interpretation of triadic recursion. Local closure cosmology shows that recursion is inherently local and horizon-bounded, yielding emergent time without requiring global renormalization structure.

3. Providing a complete reconciliation of GR and QM. The original Triadic Cosmos unified the structural roles of modern physics but did not explicitly resolve the architectural mismatch between GR and QM. Triadic closure completes this unification: GR provides geometric constraints, QM provides informational generativity, and QFT provides energetic dynamics. The full cycle closes within each closure, yielding a unified and falsifiable architecture.

5.4 A More Minimal and More General Unification

The revised unification presented in this paper is both more minimal and more general than the original Triadic Cosmos formulation:

- It requires no new physics beyond GR, QM, QFT, and thermodynamics.
- It applies to all physically realizable spacetimes, not only holographic ones.

- It explains the absence of exotic GR solutions through architectural closure.
- It resolves quantum non-locality and measurement paradoxes through horizon-bounded informational structure.
- It reframes cosmological tensions as closure-dependent rather than anomalous.

Triadic closure therefore supersedes the role previously played by AdS/CFT in the unification architecture. Holography remains a powerful illustration of geometry–information interdependence, but the fundamental mechanism of unification is local, recursive, and triadic.

5.5 Summary

The Triadic Cosmos: A Foundation for Unification provided the first architectural synthesis of modern physics. The present work completes and refines that synthesis by identifying triadic closure as the universal mechanism through which GR and QM unify. This revision removes non-essential dependencies on global holographic dualities, strengthens the dynamical interpretation of recursion, and yields a minimal, coherent, and observationally grounded unification of general relativity and quantum mechanics.

6 Illustrative Implications of Triadic Closure

Triadic closure is not merely an architectural mechanism linking GR and QM; it yields concrete, observer-dependent implications for measurement, temporal awareness, microscopic gravity, cosmological interpretation, black-hole structure, global spacetime, and quantum paradoxes. This section highlights several illustrative examples that demonstrate how triadic closure reorganizes familiar phenomena across quantum mechanics, relativity, and cosmology.

6.1 Wave-Function Collapse as Closure-Dependent

In standard quantum mechanics, wave-function collapse is treated as a non-unitary update triggered by measurement. Within triadic closure, collapse is reinterpreted as a *closure-dependent informational update*. Each observer occupies a horizon-bounded closure with its own informational capacity, causal accessibility, and recursion index. Collapse occurs when informational structure becomes embedded within the observer’s closure and cannot be further refined without violating triadic consistency.

Thus, collapse is not a universal physical event but an *observer-specific triadic update*:

- different observers may collapse the same state at different recursion depths,
- collapse reflects informational embedding rather than dynamical discontinuity,
- entanglement is preserved globally but becomes closure-specific locally.

6.2 Awareness of Time as Closure-Dependent

In GR, time is geometric; in QM, time is parametric; in triadic closure, time is emergent from recursive updates. Awareness of time therefore becomes *closure-dependent*: each observer’s temporal experience reflects the recursion index of their closure.

Consequences include:

- temporal flow is perspectival rather than universal,
- time dilation arises naturally from differences in recursion accessibility,
- cognitive time perception becomes a physical reflection of informational update rate.

6.3 Microscopic Gravity Without New Physics

A longstanding tension in physics is that QM governs extremely small scales while GR governs extremely large scales. Triadic closure resolves this mismatch by showing that both theories operate within the same architectural cycle:

$$E \rightarrow G \rightarrow I \rightarrow E.$$

At extremely small scales:

- geometric structure (G) becomes informationally saturated,
- informational capacity (I) dominates energetic evolution,
- recursion rate increases, yielding rapid emergent-time updates.

Gravity does not “break down” at small scales; rather, geometric constraints become triadically minimal. No new quantum-gravity degrees of freedom are required.

6.4 Black Holes as Triadic Closure Saturations

Black holes provide the clearest physical realization of triadic closure saturation. At the horizon:

- geometric curvature maximizes informational capacity (Bekenstein–Hawking),
- informational structure becomes holographically encoded,
- energetic structure becomes fully constrained by geometry.

Inside the horizon, recursion halts:

- emergent time ceases (no R updates),
- informational accessibility collapses to zero,
- singularities represent breakdowns of triadic closure, not physical entities.

This yields a unified interpretation of black holes across GR, QM, and thermodynamics.

6.5 No Global Spacetime: Only Closure-Dependent Spacetime

Triadic closure implies that spacetime is not a global manifold but a *closure-dependent structure*. Each observer occupies a horizon-bounded closure with its own:

- causal cone,
- informational capacity,
- recursion index,
- accessible dynamical layer.

Consequences include:

- global paradoxes (horizon problem, flatness problem) dissolve,
- cosmological parameters become closure-dependent rather than universal,
- spacetime is perspectival and emergent, not globally defined.

6.6 Heat Death Is Not a Physical End-State

The classical heat-death scenario assumes global thermodynamic equilibrium. Triadic closure rejects this assumption:

- closures have finite informational capacity and finite recursion index,
- emergent time persists as long as triadic updates occur,
- “end of time” is not a physical event but a global extrapolation.

Heat death is therefore not a physically realizable state within any closure.

6.7 The Andromeda Galaxy as Closure-Dependent Structure

For observers on Earth, the Andromeda galaxy is not a globally accessible physical object but a *closure-dependent informational structure* embedded within our horizon. Its observed properties reflect:

- the informational capacity of our cosmological closure,
- the causal accessibility of photons emitted millions of years ago,
- the recursion index of our local closure.

The predicted collision between the Milky Way and Andromeda in 4–5 billion years is not a physical event within our closure. It is a *closure-external geometric extrapolation*.

6.8 Quantum Paradoxes as Closure Artifacts

Triadic closure provides natural resolutions to several quantum paradoxes:

Schrödinger’s Cat. Superposition reflects informational structure external to the observer’s closure. Collapse occurs when the cat’s state becomes embedded within the observer’s informational capacity.

Double-Slit Experiment. Interference arises from informational structure that is not yet closure-embedded. When which-path information becomes accessible within the observer’s closure, the interference pattern disappears.

EPR and Non-Locality. Entanglement reflects informational connectivity (I) that is triadically consistent but not geometrically accessible (G) within a given closure.

6.9 Summary

These examples illustrate how triadic closure reorganizes familiar phenomena across quantum mechanics, relativity, and cosmology. Collapse, temporal awareness, microscopic gravity, black-hole structure, global spacetime, heat death, cosmological interpretation, and quantum paradoxes all become natural consequences of closure-dependent informational embedding and emergent-time recursion. Triadic closure therefore provides not only a unification of GR and QM, but a coherent interpretive framework for the full spectrum of physical phenomena.

7 Testable Predictions and Falsifiability

A unification of General Relativity and quantum mechanics must be empirically grounded and falsifiable. Triadic closure satisfies this requirement by yielding concrete, observer-dependent, horizon-bounded predictions that differ from both classical GR and standard QM. These predictions arise from the architectural constraints of the triadic cycle and from the closure-dependent nature of emergent time, informational capacity, and causal accessibility. In this section we outline several testable consequences of triadic closure and identify observational and experimental regimes capable of falsifying the framework.

7.1 Observer-Dependent Collapse and Informational Accessibility

Triadic closure predicts that wave-function collapse is not universal but *closure-dependent*. Collapse occurs when informational structure becomes embedded within an observer's horizon-bounded closure and cannot be further refined without violating triadic consistency. This yields several testable consequences:

- collapse thresholds differ for observers with different informational capacities,
- collapse timing depends on closure recursion index,
- delayed-choice and weak-measurement experiments should exhibit closure-dependent interference persistence,
- macroscopic superpositions should decohere at rates determined by closure saturation.

Experiments involving entangled photons, weak measurements, and macroscopic quantum systems (optomechanical resonators, superconducting circuits) can test these predictions.

7.2 Emergent Time and Recursion-Dependent Temporal Flow

Triadic closure predicts that temporal flow is emergent from recursive triadic updates and therefore *observer-dependent*. This yields measurable consequences:

- time dilation arises from differences in recursion accessibility rather than purely from geometric curvature,
- observers with different informational capacities should experience different effective temporal rates in cognitive-time experiments,
- quantum systems with high entanglement entropy should exhibit recursion-induced temporal slowdown relative to low-entropy systems.

These predictions can be tested through relativistic quantum clocks, entanglement-enhanced timekeeping, and cognitive-time perception experiments.

7.3 Microscopic Gravity Without New Degrees of Freedom

Triadic closure predicts that gravity at extremely small scales does not require new quantum-gravity degrees of freedom. Instead, geometric constraints become triadically minimal and informational structure dominates. This yields:

- deviations from classical Newtonian gravity at micron and submicron scales,
- absence of new particles or fields in high-energy scattering experiments,

- saturation of geometric response in high-density quantum systems,
- holographic scaling of informational capacity in microscopic gravitational regimes.

These predictions can be tested through short-range gravity experiments, atom-interferometry near massive microstructures, and high-energy collider data.

7.4 Black-Hole Interior Structure and Recursion Halt

Triadic closure predicts that recursion halts inside black-hole horizons, eliminating emergent time and informational accessibility. This yields:

- no physical singularities; singularities represent breakdowns of triadic closure,
- horizon thermodynamics should reflect recursion saturation,
- Hawking radiation spectrum should encode closure-dependent informational release,
- black-hole evaporation should exhibit recursion-index plateaus.

These predictions can be tested through gravitational-wave observations, black-hole shadow measurements, and high-precision Hawking-radiation analog experiments.

7.5 Cosmological Predictions: Closure-Dependent Parameters

Local closure cosmology predicts that cosmological parameters are not universal but *closure-dependent*. This yields:

- the Hubble tension reflects closure differences rather than new physics,
- early-universe anomalies (JWST) arise from closure-dependent recursion indices,
- cosmological “end states” such as heat death are not physically realizable within any closure,
- large-scale structure formation should exhibit closure-dependent scaling laws.

These predictions can be tested through JWST observations, precision cosmology, and comparative analysis of early- and late-time measurements.

7.6 Quantum Paradoxes as Closure Artifacts

Triadic closure predicts that quantum paradoxes arise from closure-dependent informational embedding. This yields:

- EPR correlations should exhibit closure-dependent informational accessibility,
- double-slit interference should persist until closure embedding occurs,
- macroscopic superpositions should decohere at closure-saturation thresholds,
- delayed-choice experiments should reflect recursion-index constraints.

These predictions can be tested through entanglement distribution experiments, quantum interference studies, and macroscopic superposition experiments.

7.7 Falsifiability Criteria

Triadic closure is falsifiable. It would be disproven if any of the following were observed:

- universal collapse thresholds independent of observer informational capacity,
- global spacetime structure accessible to all observers,
- new quantum-gravity degrees of freedom at microscopic scales,
- physical singularities inside black holes,
- universal cosmological parameters independent of closure,
- entanglement correlations requiring superluminal influence.

Any of these observations would violate the architectural constraints of triadic closure and falsify the unification proposed in this paper.

7.8 Summary

Triadic closure yields a rich set of testable predictions across quantum mechanics, relativity, gravity, black-hole physics, and cosmology. These predictions arise from the architectural structure of the triadic cycle and from the closure-dependent nature of emergent time, informational capacity, and causal accessibility. The framework is therefore not only a conceptual unification of GR and QM, but a falsifiable physical theory grounded in observational and experimental accessibility.

8 Limitations and Open Questions

Triadic closure provides a minimal architectural mechanism capable of reconciling General Relativity and quantum mechanics without new fields, dimensions, or dynamical extensions. However, the framework also introduces structural constraints and conceptual boundaries that define its current scope. This section outlines the principal limitations of the present formulation and identifies open questions that require further investigation.

8.1 Architectural Rather Than Dynamical Formulation

Triadic closure is an architectural mechanism, not a dynamical theory. It specifies the conditions under which geometric, energetic, and informational roles cohere, but it does not yet provide:

- explicit dynamical equations for triadic recursion,
- a closed-form mapping between recursion index and physical observables,
- a full mathematical characterization of closure boundaries.

The absence of a complete dynamical formalism limits the ability to derive precise numerical predictions. Developing such a formalism is an essential next step.

8.2 Incomplete Characterization of Recursion Index

The recursion index plays a central role in emergent time, closure-dependent causality, and observer-specific informational accessibility. However:

- its quantitative relation to geometric curvature is not fully specified,
- its evolution across cosmological scales remains partially understood,
- its role in black-hole evaporation and horizon thermodynamics requires refinement.

A more precise definition of recursion index would strengthen the predictive power of the framework.

8.3 Local Closure Without Global Completion

Triadic closure is inherently local and horizon-bounded. While this resolves many global paradoxes, it also raises open questions:

- Can closures be consistently patched together into a global manifold?
- Are closure transitions possible, and if so, under what conditions?
- How does closure-dependent spacetime relate to global cosmological evolution?

The relationship between local closures and global spacetime structure remains an open architectural question.

8.4 Black-Hole Interiors and Recursion Halt

The framework predicts that recursion halts inside black-hole horizons, eliminating emergent time and informational accessibility. This interpretation avoids singularities but raises further questions:

- What is the precise mathematical structure of a recursion-halted region?
- How does halted recursion interact with Hawking radiation and information release?
- Can recursion resume under extreme conditions, or is halt absolute?

These questions are central to understanding the full triadic structure of black holes.

8.5 Microscopic Gravity and Experimental Accessibility

Triadic closure predicts that gravity at extremely small scales is triadically minimal rather than requiring new quantum-gravity degrees of freedom. However:

- current experimental sensitivity at micron and submicron scales is limited,
- deviations from Newtonian gravity may be subtle and difficult to isolate,
- informational saturation effects may require novel experimental setups.

Improved short-range gravity experiments are necessary to test these predictions.

8.6 Cosmological Closure and Parameter Variability

Local closure cosmology reframes cosmological parameters as closure-dependent rather than universal. This raises several open questions:

- How do recursion indices evolve across cosmic time?
- Can closure-dependent cosmological parameters be inferred from observational data?
- What is the relationship between closure transitions and large-scale structure?

A full cosmological model based on triadic closure remains to be developed.

8.7 Quantum Paradoxes and Informational Embedding

Triadic closure resolves quantum paradoxes through closure-dependent informational embedding. However:

- the precise boundary between closure-external and closure-internal information is not yet mathematically formalized,
- the role of recursion index in entanglement accessibility requires further study,
- the relationship between informational embedding and decoherence remains partly conceptual.

A more rigorous treatment of informational boundaries would strengthen the interpretive framework.

8.8 Summary

Triadic closure provides a minimal and coherent architectural unification of GR and QM, but it remains an early-stage structural framework. Key limitations include the absence of a complete dynamical formalism, incomplete characterization of recursion index, and open questions regarding black-hole interiors, microscopic gravity, cosmological closure, and informational embedding. Addressing these limitations will be essential for developing triadic closure into a fully predictive and mathematically complete physical theory.

9 Conclusion

This paper has presented triadic closure as a minimal and falsifiable architectural mechanism capable of reconciling General Relativity and quantum mechanics without introducing new fields, dimensions, symmetries, or dynamical extensions. Building on the structural ontology developed in the Triadic Cosmos ecosystem, we have shown that energy, geometry, and information form three irreducible roles whose closed interdependence yields emergent time, observer-dependent causality, horizon-bounded accessibility, and a unified account of physical law.

Triadic closure resolves the longstanding incompatibility between GR and QM by revealing that their apparent conflict arises not from missing physics, but from the absence of an architectural grammar linking their roles. GR describes the geometric constraints that bound informational capacity; QM describes the informational structure that generates energetic evolution; thermodynamics and QFT describe the energetic dynamics that shape geometry. Within each closure, these roles cohere through the triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow E,$$

yielding a unified and perspectival physical framework.

The implications of this architecture are broad. Wave-function collapse becomes closure-dependent rather than universal; temporal awareness becomes a reflection of recursion index; microscopic gravity requires no new degrees of freedom; black holes become triadic closure saturations rather than singularities; global spacetime dissolves into closure-dependent causal structure; cosmological parameters become horizon-specific rather than universal; and classical quantum paradoxes emerge as artifacts of closure-external informational structure. These consequences demonstrate that triadic closure is not merely a conceptual synthesis, but a physically meaningful reorganization of the foundations of modern physics.

Triadic closure is also falsifiable. It yields testable predictions across quantum mechanics, relativity, black-hole physics, cosmology, and cognitive-time perception. The framework would be disproven by universal collapse thresholds, global spacetime accessibility, new quantum-gravity degrees of freedom, physical singularities, closure-independent cosmological parameters, or entanglement correlations requiring superluminal influence. This falsifiability distinguishes triadic closure from many existing unification programs.

At the same time, the framework has limitations. A complete dynamical formalism for triadic recursion remains to be developed; the quantitative structure of recursion index requires refinement; the relationship between local closures and global cosmological evolution remains an open question; and the mathematical characterization of informational boundaries must be strengthened. These limitations define a clear research agenda for future work.

Overall, triadic closure provides a coherent, minimal, and observationally grounded unification of GR and QM. It shows that modern physics already contains all necessary ingredients for unification, and that the fragmentation between its major theories arises from architectural incompleteness rather than physical inconsistency. By supplying the missing architectural grammar linking energy, geometry, and information, triadic closure positions the Triadic Cosmos as a complete and falsifiable foundation for a unified description of physical reality.

A Comparison With Existing Unification Frameworks

This appendix compares triadic closure with major contemporary unification programs, including string theory, loop quantum gravity (LQG), and holographic dualities such as AdS/CFT. The goal is not to provide an exhaustive review, but to highlight structural differences in assumptions, ontology, and explanatory scope.

A.1 Comparison Table

Framework	Core Ontology	Unification Mechanism	Limitations / Assumptions
String Theory	1D strings, higher-dimensional manifolds, supersymmetry	Unifies forces via vibrational modes of strings; geometry emerges from compactification	Requires extra dimensions; relies on supersymmetry; landscape problem; no observer-dependence
Loop Quantum Gravity	Spin networks, discrete geometry, quantized area/volume	Quantizes space-time directly; GR becomes a quantum theory of geometry	No unification with SM; limited dynamical predictions; no informational role; no emergent time
AdS/CFT and Holography	Bulk geometry dual to boundary CFT; entanglement versus geometry	Unifies gravity and QFT via duality; geometry emerges from entanglement	Requires AdS boundary; global conformal symmetry; not applicable to our universe; duality not universal
Triadic Closure	Energy, geometry, information as irreducible roles; horizon-bounded closures	Unifies GR and QM via architectural cycle $E \rightarrow G \rightarrow I \rightarrow E$; time emerges from recursion	No extra dimensions; no new fields; requires closure structure; dynamical formalism still incomplete

A.2 Discussion

Triadic closure differs from existing unification programs in several fundamental ways:

- 1. Minimal Ontology.** String theory and LQG introduce new fundamental entities (strings, spin networks, discrete geometry). Triadic closure introduces no new physical objects; it reorganizes existing physics through architectural roles.
- 2. No Extra Dimensions or Global Boundaries.** String theory requires compactified dimensions; holography requires a global AdS boundary. Triadic closure is fully local and horizon-bounded, applicable to any physically realized spacetime.

3. Information as a Structural Role. Triadic closure treats informational structure as irreducible and dynamically essential. Neither string theory nor LQG incorporate information as a fundamental physical role.

4. Emergent Time Through Recursion. Time emerges from recursive triadic updates. This differs from:

- string theory (time is geometric),
- LQG (time is quantized or relational),
- holography (time is boundary-parametric).

5. Observer-Dependent Closures. Triadic closure predicts perspectival spacetime, collapse, and temporal flow. Existing frameworks assume global spacetime or global Hilbert-space structure.

6. No Singularities. Black-hole interiors become recursion-halted architectural gaps, not physical singularities. String theory and LQG attempt to resolve singularities dynamically; triadic closure resolves them architecturally.

7. Falsifiability. Triadic closure yields explicit falsifiability criteria (observer-dependent collapse thresholds, closure-dependent cosmological parameters, absence of new quantum-gravity particles). String theory and LQG have limited falsifiability due to high-energy inaccessibility.

A.3 Summary

Triadic closure provides a minimal, local, and perspectival unification of GR and QM that differs structurally from string theory, LQG, and holography. It requires no new fields, no extra dimensions, and no global dualities. Instead, it reorganizes existing physics through the interdependence of energy, geometry, and information within horizon-bounded closures.

B Core Formalism of Triadic Closure

This appendix presents the minimal mathematical formalism underlying triadic closure. The formalism is architectural rather than dynamical: it specifies the structural roles, constraints, and update relations required for physical realizability. It provides the foundation for the unification of General Relativity and quantum mechanics developed in the main text.

B.1 Triadic State Space

A physical system is represented by a triadic state

$$X = (E, G, I),$$

where:

- E denotes energetic structure (stress–energy, dynamical resources),
- G denotes geometric structure (curvature, causal connectivity),
- I denotes informational structure (entropy, entanglement, distinguishability).

These roles are irreducible: none can be derived from the others, and each constrains the others through the triadic cycle.

B.2 Triadic Cycle

The interdependence of the triadic roles is expressed through the cycle

$$E \rightarrow G \rightarrow I \rightarrow E.$$

Each leg of the cycle represents a structural dependency:

$E \rightarrow G$	energetic structure shapes geometric constraints,
$G \rightarrow I$	geometric structure bounds informational capacity,
$I \rightarrow E$	informational structure generates energetic evolution.

The cycle must close for physical realizability.

B.3 Recursive Update Operator

Dynamics arise from the recursive update operator

$$X_{n+1} = R(X_n),$$

where n indexes emergent-time steps. The operator decomposes into three structural update functions:

$$\begin{aligned} E_{n+1} &= f(G_n, I_n), \\ G_{n+1} &= g(E_n, I_n), \\ I_{n+1} &= h(E_n, G_n). \end{aligned}$$

The functions f , g , and h are not dynamical laws but architectural constraints: they specify how each role reorganizes the others across recursion.

B.4 Closure Definition

A *closure* $\mathcal{C}$ is a horizon-bounded region of spacetime in which the triadic cycle is well-defined and recursion is possible. Formally, a closure satisfies:

1. **Positivity:** $E > 0$, $G > 0$, $I > 0$.
2. **Holographic consistency:** informational capacity scales with horizon area.
3. **Causal accessibility:** energetic and geometric roles satisfy local causal bounds.
4. **Recursive stability:** R is well-defined for all $X \in \mathcal{C}$.
5. **Cycle closure:** $E \rightarrow G \rightarrow I \rightarrow E$ completes without contradiction.

If any condition fails, recursion halts and emergent time ceases.

B.5 Emergent Time

Emergent time is defined as the ordering of recursive triadic states:

$$t \equiv n \quad \text{where} \quad X_{n+1} = R(X_n).$$

Time is therefore:

- perspectival (closure-dependent),
- discrete at the architectural level,
- continuous only in the coarse-grained limit,
- absent in recursion-halted regions (e.g., black-hole interiors).

B.6 Observer-Specific Closures

Each observer occupies a distinct closure $\mathcal{C}_{\text{obs}}$ with its own:

- informational capacity I_{obs} ,
- causal cone G_{obs} ,
- recursion index n_{obs} ,
- accessible dynamical layer E_{obs} .

Closures may overlap but never coincide exactly. Collapse, temporal awareness, and causal structure are therefore observer-dependent.

B.7 Black Holes as Recursion-Halted Regions

A black-hole interior is characterized by:

$$R(X) \quad \text{undefined.}$$

At the horizon:

- I saturates (holographic bound),
- recursion halts (n ceases to advance),
- emergent time disappears,
- causal accessibility collapses.

Thus black holes are architectural gaps in closure rather than physical volumes.

B.8 Global Spacetime as Closure-External

Global spacetime manifolds are mathematical constructs. Physical reality consists of closure-dependent spacetime patches $\mathcal{C}$ with perspectival causal structure. No observer has access to global spacetime; only closure-external geometric extrapolations exist.

B.9 Summary

The core formalism defines the triadic roles, their interdependence, the recursive update operator, and the structural constraints required for closure. It provides the mathematical foundation for emergent time, observer-dependent causality, closure-specific collapse, and the unification of GR and QM developed in the main text.

C Worked Example: Triadic Closure in a Double-Slit Scenario

This appendix presents a minimal worked example of triadic closure in a double-slit-like quantum interference setup. The goal is not to derive full experimental predictions, but to illustrate how the triadic roles, recursive updates, and closure-dependent collapse operate in a concrete scenario.

C.1 Setup

Consider a single-photon source, a double-slit barrier, and a detection screen. An observer $\mathcal{O}$ is located behind the screen and occupies a closure $\mathcal{C}_{\mathcal{O}}$ with:

- informational capacity $I_{\mathcal{O}}$,
- causal cone $G_{\mathcal{O}}$,
- recursion index $n_{\mathcal{O}}$,
- accessible dynamical layer $E_{\mathcal{O}}$.

The photon's path from source to screen is represented by a triadic state

$$X = (E, G, I),$$

where:

- E encodes the photon's energy and its coupling to the apparatus,
- G encodes the geometric configuration (source, slits, screen),
- I encodes the informational structure (path superposition, phase relations).

C.2 Triadic Evolution Without Which-Path Information

Before any which-path information is accessible to $\mathcal{O}$, the triadic state evolves under recursion:

$$X_{n+1} = R(X_n),$$

with:

$$\begin{aligned} E_{n+1} &= f(G_n, I_n), \\ G_{n+1} &= g(E_n, I_n), \\ I_{n+1} &= h(E_n, G_n). \end{aligned}$$

In this regime:

- I_n encodes a coherent superposition of paths through both slits,
- G_n encodes a geometric configuration that supports interference,
- E_n encodes the photon's energy and its coupling to the screen.

The observer's closure $\mathcal{C}_{\mathcal{O}}$ does not yet contain path-resolved information. Interference arises because I_n remains closure-external: the full informational structure is not embedded within $I_{\mathcal{O}}$.

C.3 Closure-Dependent Collapse With Which-Path Information

Now suppose a which-path detector is introduced, and its output becomes accessible to $\mathcal{O}$. The observer's closure $\mathcal{C}_{\mathcal{O}}$ expands informationally:

$$I_{\mathcal{O}} \rightarrow I'_{\mathcal{O}} > I_{\mathcal{O}},$$

such that path information becomes closure-internal.

Triadic closure predicts that collapse occurs when:

$$I_n \subseteq I'_{\mathcal{O}},$$

and further refinement of I_n would violate triadic consistency (e.g., exceed holographic capacity or causal accessibility). At this point:

- the superposition in I_n is replaced by a path-resolved state,
- G_n reorganizes to reflect a non-interfering geometric configuration,
- E_n updates to encode a classical-like detection event.

Interference disappears not because the photon “chooses” a path, but because the informational structure becomes closure-embedded and the triadic cycle reorganizes.

C.4 Observer-Dependence

Consider a second observer $\mathcal{O}'$ who does not have access to the which-path detector output. Their closure $\mathcal{C}_{\mathcal{O}'}$ satisfies:

$$I_{\mathcal{O}'} < I'_{\mathcal{O}},$$

and:

$$I_n \not\subseteq I_{\mathcal{O}'}.$$

For $\mathcal{O}'$, the informational structure remains closure-external, and interference persists. Collapse is therefore *observer-dependent*:

- $\mathcal{O}$ observes a non-interfering pattern,
- $\mathcal{O}'$ observes an interfering pattern,
- both are triadically consistent within their respective closures.

This illustrates how triadic closure reconciles perspectival collapse without modifying quantum mechanics.

C.5 Recursion Index and Temporal Ordering

The recursion index n provides an emergent-time ordering of triadic states:

$$X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots .$$

For $\mathcal{O}$:

- the introduction of which-path information increases recursion complexity,
- collapse occurs at a specific recursion step n_{collapse} ,
- temporal awareness reflects the ordering of updates in $\mathcal{C}_{\mathcal{O}}$.

For $\mathcal{O}'$:

- recursion proceeds without path-resolved embedding,
- interference persists across multiple recursion steps,
- temporal awareness reflects a different update sequence.

Emergent time is therefore closure-dependent and perspectival.

C.6 Summary

This worked example shows how triadic closure operates in a double-slit-like scenario:

- interference arises when informational structure is closure-external,
- collapse occurs when informational structure becomes closure-internal,
- different observers inhabit different closures and therefore experience different collapse behavior,
- emergent time reflects the ordering of triadic updates within each closure.

The example illustrates the core claims of the main text in a concrete setting, without requiring modifications to standard quantum mechanics or General Relativity.

D Worked Example: Triadic Closure in a Black-Hole Scenario

This appendix presents a concrete worked example illustrating how triadic closure operates in the context of a black hole. Conceptual background on closure structure, recursion halt, and architectural gaps is provided in Appendix E. Here we focus on the operational behavior of the triadic cycle in a specific physical configuration.

D.1 Setup

Consider a Schwarzschild-like black hole with mass M and horizon radius

$$r_H = \frac{2GM}{c^2}.$$

An external observer $\mathcal{O}$ remains at radius $r \gg r_H$ and occupies a closure $\mathcal{C}_{\mathcal{O}}$ with triadic roles $(E_{\mathcal{O}}, G_{\mathcal{O}}, I_{\mathcal{O}})$.

The black-hole region is represented by a triadic state

$$X_{\text{BH}} = (E_{\text{BH}}, G_{\text{BH}}, I_{\text{BH}}),$$

encoding exterior curvature, horizon geometry, and holographic informational capacity.

D.2 Horizon as Operational Boundary

For $\mathcal{O}$, the horizon at r_H acts as an operational boundary:

- G_{BH} restricts causal accessibility,
- I_{BH} saturates at the holographic bound,
- E_{BH} couples to exterior dynamics through Hawking radiation.

The observer's closure includes $r > r_H$ but not the interior region.

D.3 Recursion Behavior

Triadic recursion proceeds normally in the exterior:

$$X_{n+1}^{\text{ext}} = R(X_n^{\text{ext}}),$$

where exterior triadic roles update through energetic inflow, geometric curvature, and informational accessibility.

Inside the horizon, recursion halts:

$$R(X_{\text{BH}}) \text{ undefined for } r < r_{\text{H}}.$$

This operational halt is explained conceptually in Appendix E.

D.4 Exterior Dynamics and Evaporation

Hawking radiation appears as:

- a reorganization of E_{BH} into E^{ext} ,
- a release of I_{BH} into I^{ext} ,
- a gradual update of G^{ext} as M decreases.

Evaporation is therefore an exterior recursion process.

D.5 Observer-Specific Differences

A second observer $\mathcal{O}'$ closer to the horizon occupies a different closure $\mathcal{C}_{\mathcal{O}'}$ with:

- different causal accessibility,
- different informational capacity,
- different recursion index.

Both observers experience triadically consistent but perspectively different exterior dynamics.

D.6 Summary

This example shows how triadic closure governs black-hole behavior at the operational level: exterior recursion, horizon-defined boundaries, evaporation as triadic reorganization, and observer-dependent accessibility. Conceptual interpretation is provided in Appendix E.

E Closure Structure: Black Holes and Observer-Specific Closures

This appendix provides the conceptual background for closure structure in triadic closure, including the architectural interpretation of black holes and the perspectival nature of observer-specific closures. Operational behavior in a concrete black-hole scenario is given in Appendix D; here we focus exclusively on the structural and interpretive aspects.

E.1 Closures as Horizon-Bounded Physical Domains

A closure $\mathcal{C}$ is defined as a horizon-bounded region in which the triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow E$$

is well-defined and recursive updates are possible. Each closure has:

- **informational capacity** $I_{\mathcal{C}}$ bounded by its horizon,
- **geometric accessibility** $G_{\mathcal{C}}$ defining causal structure,
- **energetic accessibility** $E_{\mathcal{C}}$ defining dynamical resources,
- **recursion index** $n_{\mathcal{C}}$ defining emergent-time flow.

Closures are inherently local and perspectival: no observer has access to global spacetime, only to the horizon-bounded region defined by their closure.

E.2 Black Holes as Architectural Gaps in Closure

In General Relativity, black holes are geometric regions bounded by an event horizon. Within triadic closure, their interpretation becomes architectural: black holes correspond to *internal gaps in closure*. At the horizon:

- informational capacity saturates at the holographic bound,
- causal accessibility collapses,
- recursion halts,
- emergent time ceases.

Inside the horizon, the triadic cycle cannot close:

$$R(X) \quad \text{undefined for interior states.}$$

Thus, black holes are not physical volumes within spacetime but *regions where closure internally fails to exist*. They are architectural voids rather than dynamical domains. This interpretation unifies:

- horizon thermodynamics,
- holographic entropy,
- GR curvature,
- quantum informational bounds,

within a single structural mechanism.

Absence of Black Holes Within Our Closure. The nearest confirmed black-hole candidate (e.g., Gaia BH1 at ~ 1.6 kpc) lies far outside the causal and informational accessibility of any human observer. Within triadic closure, this implies that our local closure $\mathcal{C}_{\text{human}}$ contains no physically accessible black holes. Black holes exist as closure-external geometric structures, but not as triadically embedded regions within our observer-specific closure.

This reinforces the perspectival nature of closure and the fundamentally local character of triadic architecture.

E.3 Observer-Specific Closures

Triadic closure predicts that each observer occupies a distinct closure $\mathcal{C}_{\text{obs}}$ with its own:

- informational capacity I_{obs} ,
- causal cone G_{obs} ,
- recursion index n_{obs} ,
- accessible dynamical layer E_{obs} .

Closures may overlap but never coincide exactly. Overlap allows:

- shared perception of objects,
- synchronized causal accessibility,
- partial alignment of recursion indices,
- mutual energetic and informational influence.

However, differences in cognitive processing, informational embedding, causal accessibility, and recursion rate ensure that each observer experiences a slightly different emergent-time flow and informational landscape.

This perspectival structure explains:

- observer-dependent collapse thresholds,
- variations in temporal awareness,
- differences in perceptual resolution,
- perspectival interpretations of cosmological phenomena.

E.4 Relation to Worked Examples

Appendix D provides a concrete operational example of how these architectural principles manifest in a black-hole scenario. The present appendix supplies the conceptual foundation for that example.

E.5 Summary

Black holes become architectural gaps in closure rather than physical regions of spacetime, and individual observers inhabit distinct but overlapping closures with their own informational and causal structure. These consequences reinforce the local, perspectival nature of triadic closure and clarify how GR and QM unify within horizon-bounded dynamical niches.

F Minimal Instantiation of Triadic Update Functions and Numerical Example

This appendix provides (i) a minimal concrete instantiation of the triadic update functions f , g , and h introduced in Appendix A, and (ii) a numerical example illustrating closure capacity using the horizon entropy of a Schwarzschild black hole. The goal is not to propose final dynamical laws, but to demonstrate that the triadic architecture admits simple, physically meaningful instantiations.

F.1 Minimal Instantiation of the Triadic Update Functions

Recall the recursive update operator:

$$X_{n+1} = R(X_n), \quad X_n = (E_n, G_n, I_n).$$

A minimal, physically motivated instantiation of the update functions is:

$$E_{n+1} = E_n + \alpha \Delta I_n, \tag{1}$$

$$G_{n+1} = G_n + \beta \Delta E_n, \tag{2}$$

$$I_{n+1} = I_n + \gamma \Delta G_n, \tag{3}$$

where:

- ΔI_n is the change in accessible informational structure during step n ,
- ΔE_n is the change in energetic structure,
- ΔG_n is the change in geometric structure,
- α, β, γ are small, closure-dependent coupling coefficients.

Equations (1)–(3) express the triadic cycle:

$$I \rightarrow E, \quad E \rightarrow G, \quad G \rightarrow I,$$

in a minimal algebraic form. These update rules satisfy:

- **architectural consistency:** each role reorganizes the next,
- **closure dependence:** α, β, γ vary with horizon size,
- **emergent time:** recursion index n defines temporal ordering,
- **holographic bounds:** I_{n+1} cannot exceed closure capacity.

This instantiation is not intended as a final physical model, but as a demonstration that the triadic architecture admits simple, interpretable update rules.

F.2 Numerical Example: Horizon Entropy of a Schwarzschild Black Hole

To illustrate closure capacity numerically, consider the horizon entropy of a Schwarzschild black hole of mass

$$M = 10M_\odot,$$

where $M_\odot$ is the solar mass.

Horizon Radius. The Schwarzschild radius is:

$$r_H = \frac{2GM}{c^2} = \frac{2G(10M_\odot)}{c^2} \approx 29.5 \text{ km}.$$

Horizon Area.

$$A = 4\pi r_H^2 \approx 4\pi(2.95 \times 10^4 \text{ m})^2 \approx 1.1 \times 10^{10} \text{ m}^2.$$

Horizon Entropy. The Bekenstein–Hawking entropy is:

$$S_{\text{BH}} = \frac{k_{\text{B}} c^3 A}{4G\hbar} \approx 1.5 \times 10^{79} k_{\text{B}}.$$

This value represents the maximal informational capacity I_{H} of the horizon. Within triadic closure:

- I_{H} is the upper bound for I_n in any closure containing the horizon,
- recursion halts when I_n reaches I_{H} ,
- the interior region has no informational capacity ($I = 0$),
- evaporation corresponds to a decrease in I_{H} as A shrinks.

Thus, the numerical value of S_{BH} provides a concrete example of closure capacity and demonstrates how geometric structure (G) bounds informational structure (I) in the triadic architecture.

F.3 Summary

This appendix provides:

- a minimal algebraic instantiation of the triadic update functions,
- a numerical example of closure capacity using black-hole horizon entropy,
- a demonstration of how geometric, energetic, and informational roles interact quantitatively within triadic closure.

These elements complement the architectural formalism of Appendix A and the operational examples of Appendices B and C.

G Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: <https://github.com/triadic-cosmos>.

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